

Day 5: Data visualization (cont.)

Lecturer: Msc. Minh Tan Le

Outlines

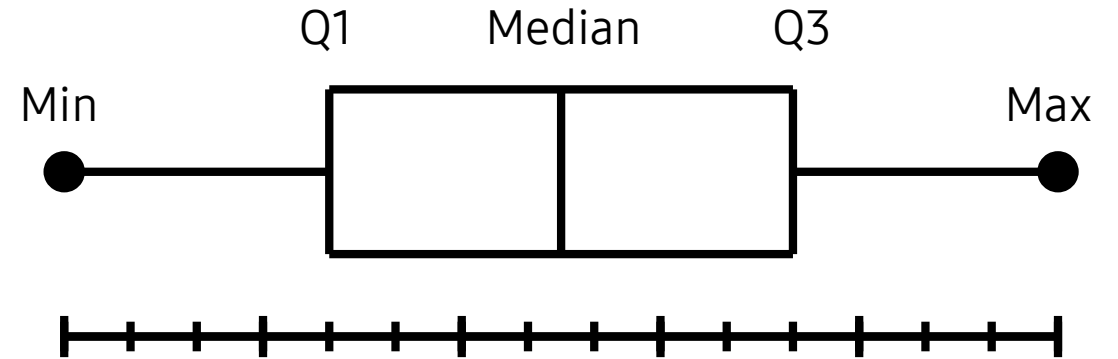
I. Advanced diagrams

- Box diagram & candlestick chart
- Network diagram

II. Report vs. Dashboard

Box diagram

- Display:
 - Statistics moments
 - Median, Q1, Q3
 - Limits
 - Min, max



Steps

1. Sort the data in ascending order.
2. Find median.
3. Find Quatile values.
 - Q1: Local median from min to (& include) global median.
 - Q3: Local median from right after (& include) median to max.
4. Determine summary numbers
 - Min, Q1, median, Q3, max.
5. Draw

34, 25, 29, 38, 29, 30, 35, 37, 35, 28 (10)

34, 25, 29, 38, 29, 30, 35, 37, 35, 28



25, 28, 29, 29, 30, 34, 35, 35, 37, 38

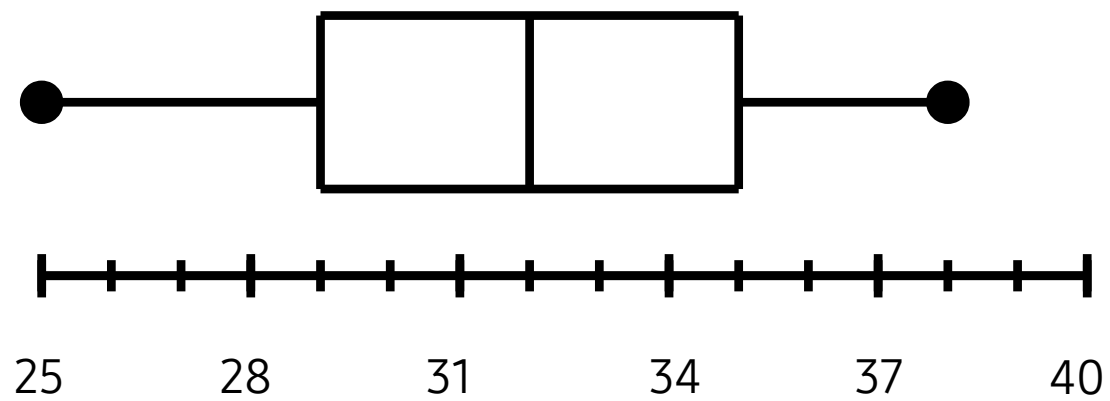
$$(30 + 34)/2 = 32$$

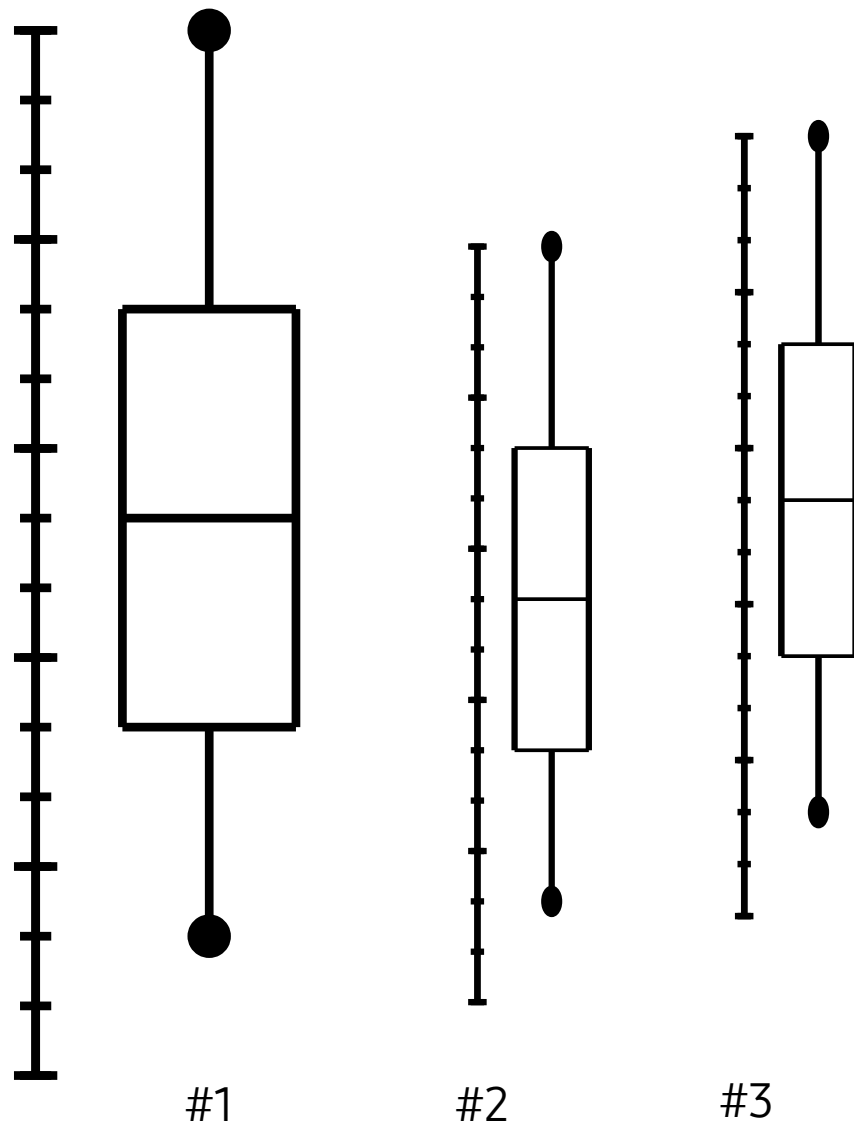


25, 28, 29, 29, **30**, **34**, 35, 35, 37, 38

[25, 28, **29**, 29, 30], [34, 35, **35**, 37, 38]

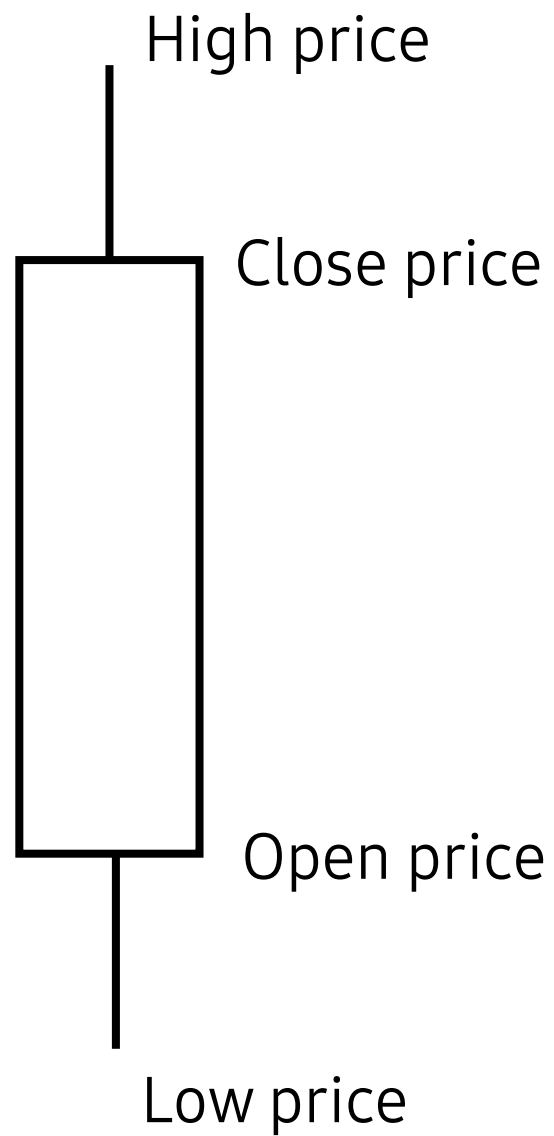
25, 29, 32, 35, 38

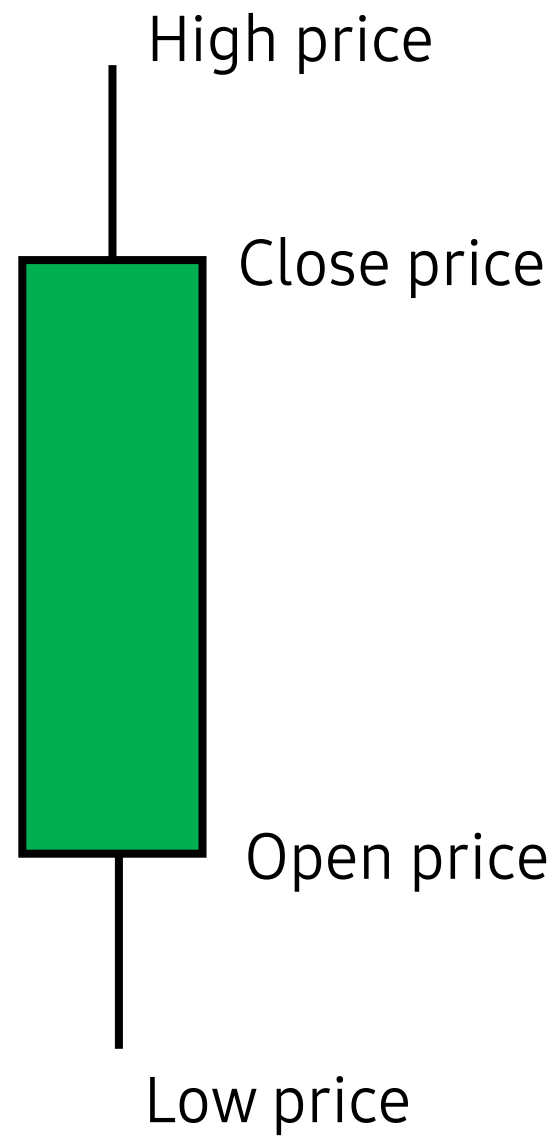
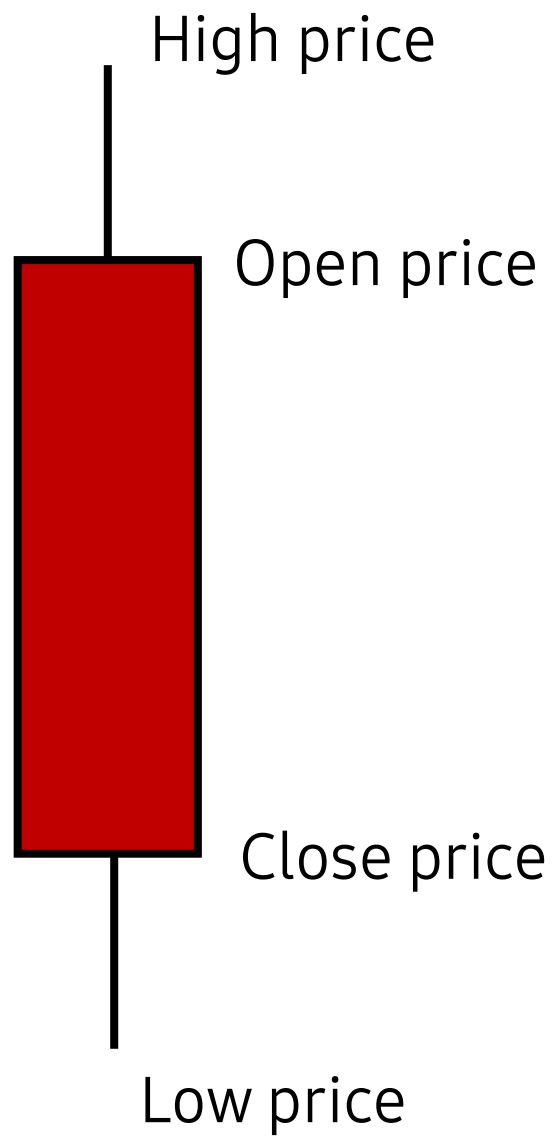




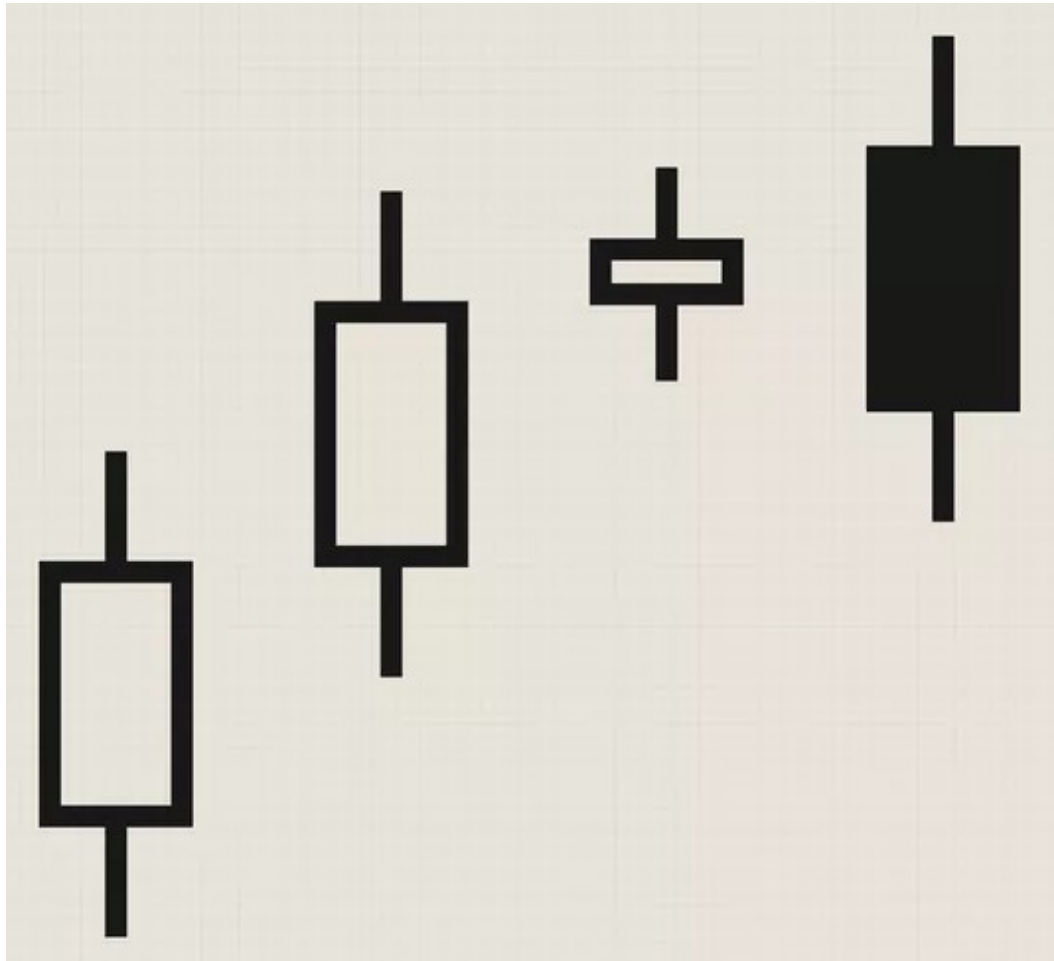
Experiments

- In 1700s, a Japanese named Homma discovered the relationships between prices and supplies/demands of rice.
 - Price is affected by traders' psychology.
- Candlestick chart can illustrate that psychology through value amplitude, and support temporary trending prediction.

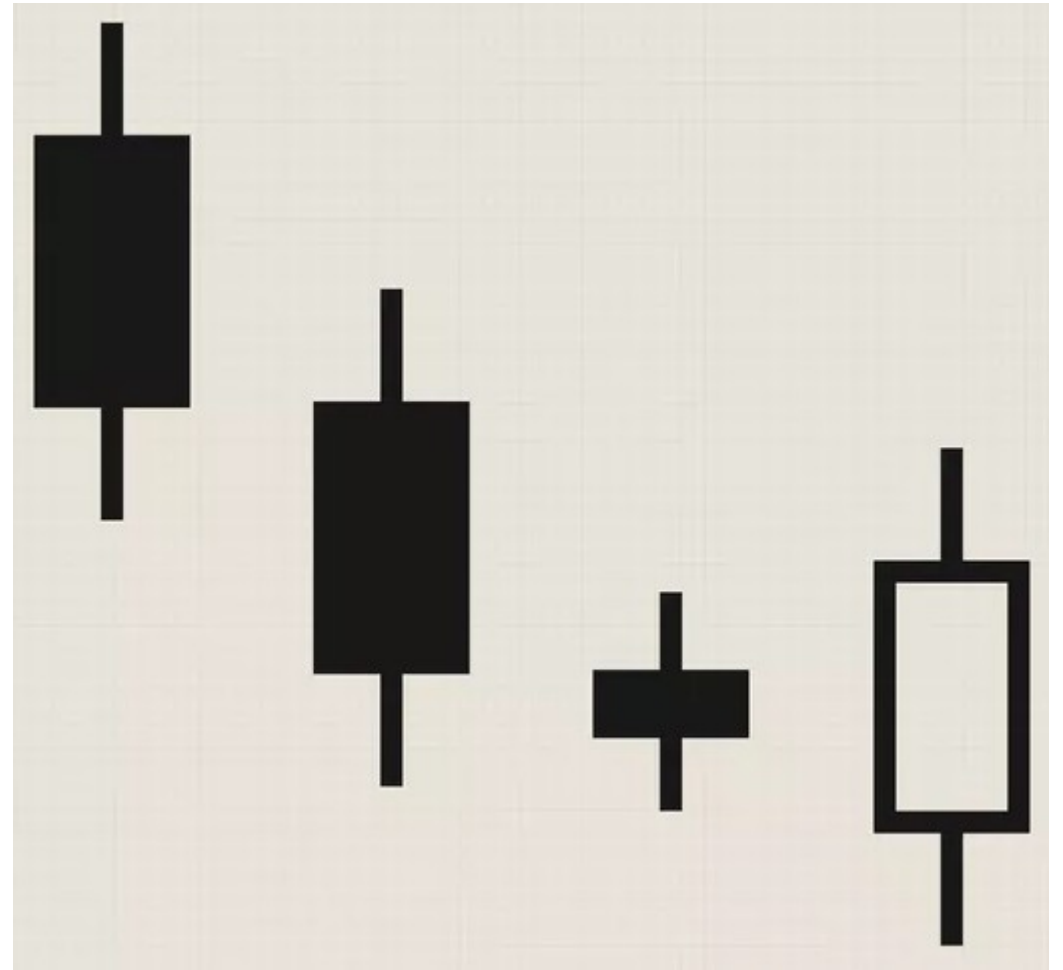






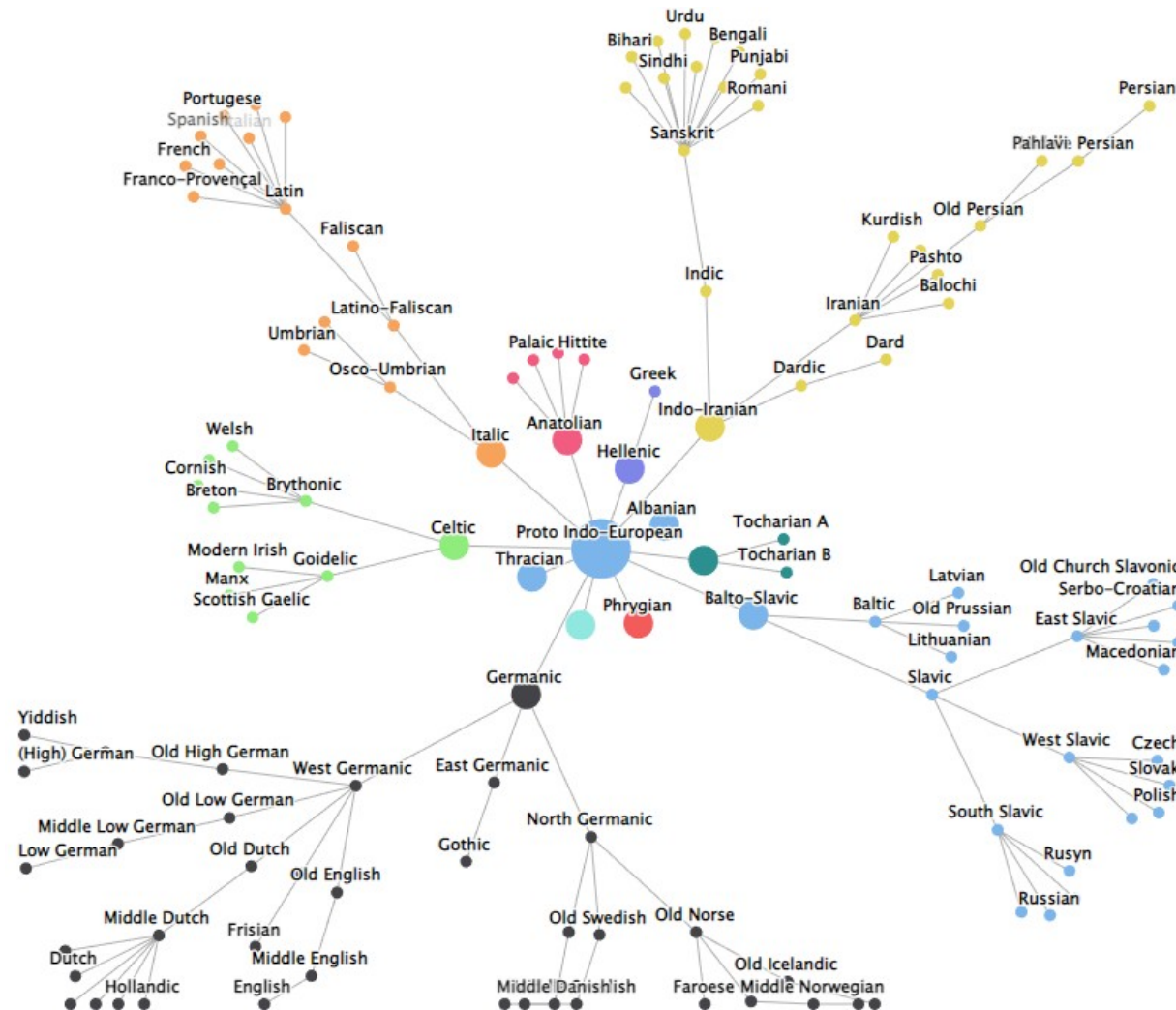


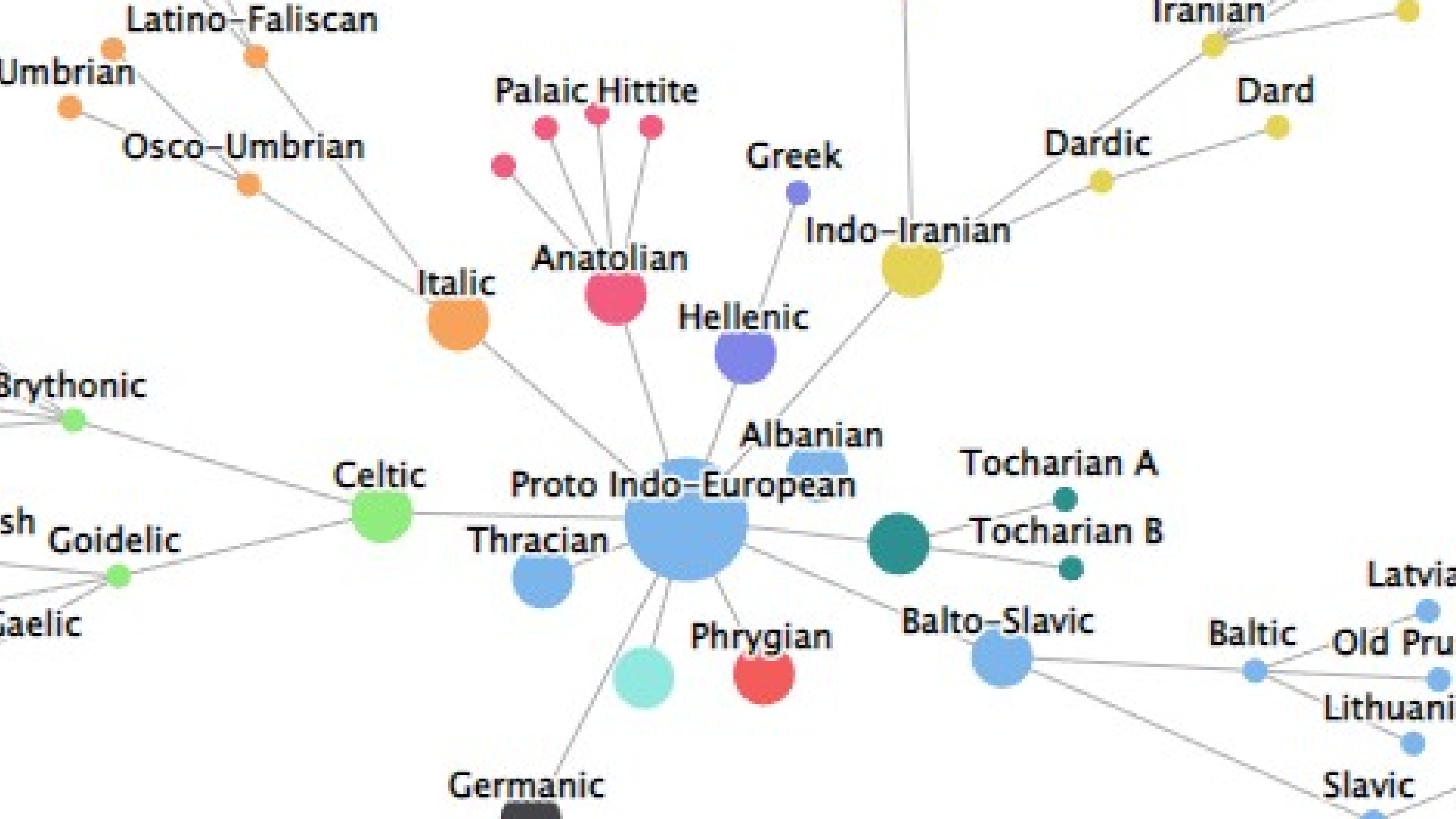
Bearish Engulfing Pattern

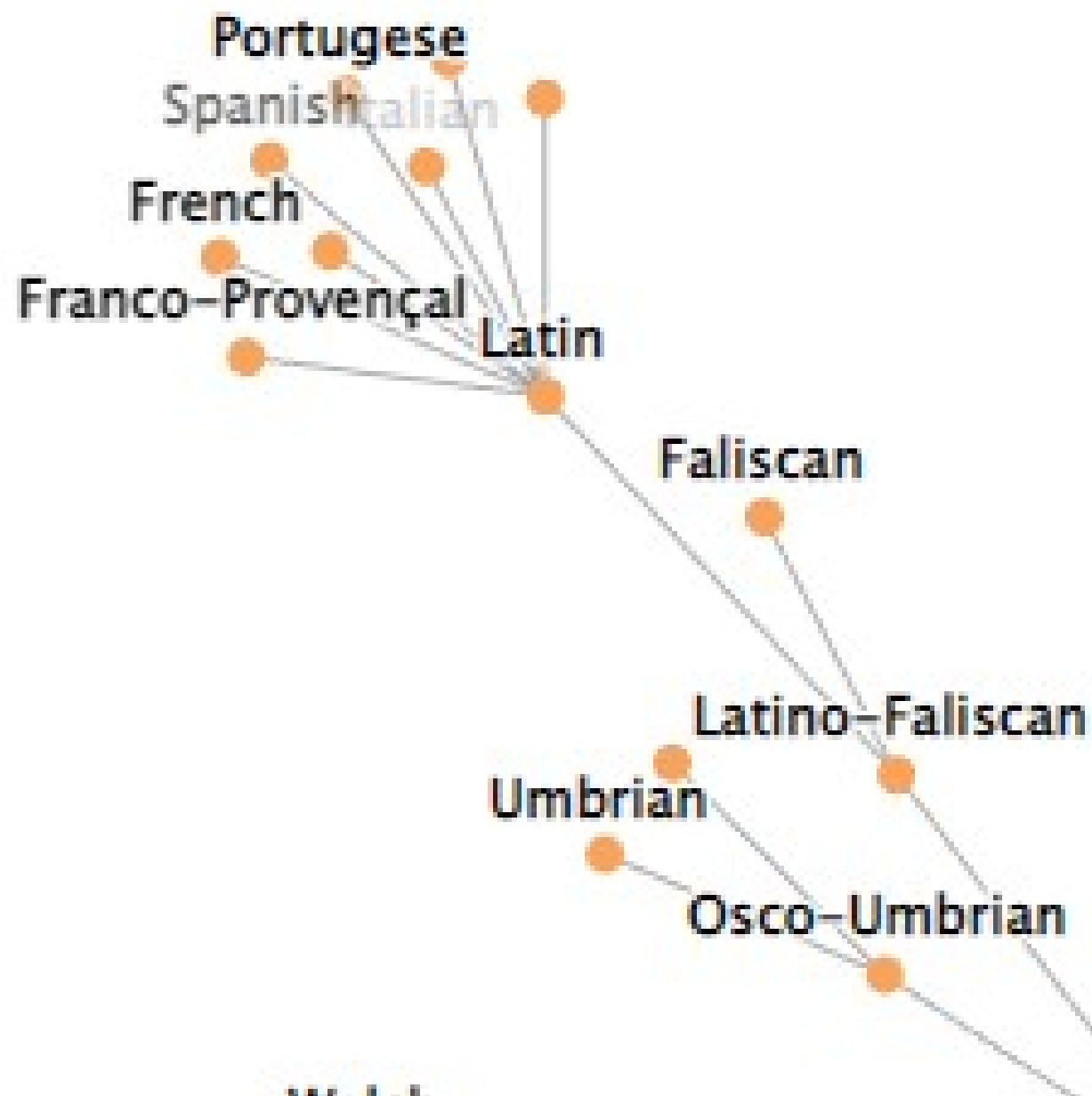


Bullish Engulfing Pattern

Network diagram







Tools

- NetworkX
- Graphviz
- Zeppelin Network

Use case

Draw a network diagram to show the correlation between two main parties from the 104th US house & senate votes.

Materials

- congress_votes_104-1995:

<https://cloud.cntt.io/s/dPSLTE42txim97p>

- legislators-historical: <https://cloud.cntt.io/s/rbGpgpxeJb2tWDk>

```
- id:
  bioguide: B000226
  govtrack: 401222
  icpsr: 507
  wikipedia: Richard Bassett (Delaware politician)
  wikidata: Q518823
  google_entity_id: kg:/m/02pz46
  name:
    first: Richard
    last: Bassett
  bio:
    birthday: '1745-04-02'
    gender: M
  terms:
    - type: sen
      start: '1789-03-04'
      end: '1793-03-03'
      state: DE
  class: 2
  party: Anti-Administration
```

legislators-historical.yaml

```
Senate Vote #247 1995-06-09T11:03:00 - On the Amendment S.Amdt. 12
person,state,district,vote,name,party
300001,HI,,Yea,"Sen. Daniel Akaka [D, 1990-2012]",Democrat
300005,MT,,Yea,"Sen. Max Baucus [D, 1978-2014]",Democrat
300007,UT,,Yea,"Sen. Robert Bennett [R, 1993-2010]",Republican
300008,DE,,Not Voting,"Sen. Joseph Biden [D, 1973-2009]",Democrat
300009,NM,,Yea,"Sen. Jeff Bingaman [D, 1983-2012]",Democrat
```

Vote data by bill, year, id of legislators

References

- https://www.senate.gov/legislative/LIS/roll_call_lists/vote_memo_104_1.htm
- <https://www.govtrack.us/congress/votes/104-1995/s247>

On the Amendment S.Amdt. 1267 to S. 652 (Telecommunications Act of 1996)

Jun 9, 1996 at 11:03 a.m. ET.

This was a vote to approve or reject an amendment to [S. 652 \(104th\)](#) in the Senate.

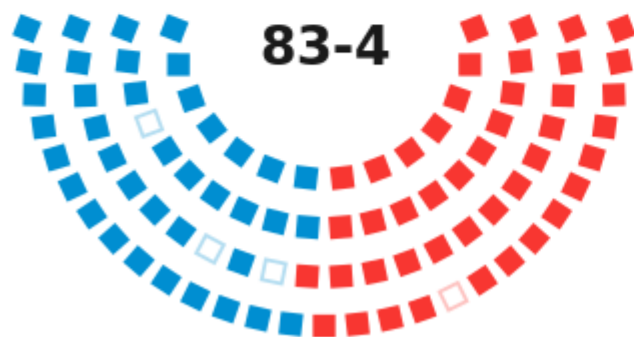
Totals

	All Votes	Republicans	Democrats
Yea	95 %  83	 44	 39
Nay	5 %  4	 1	 3
Not Voting	 13	 9	 4

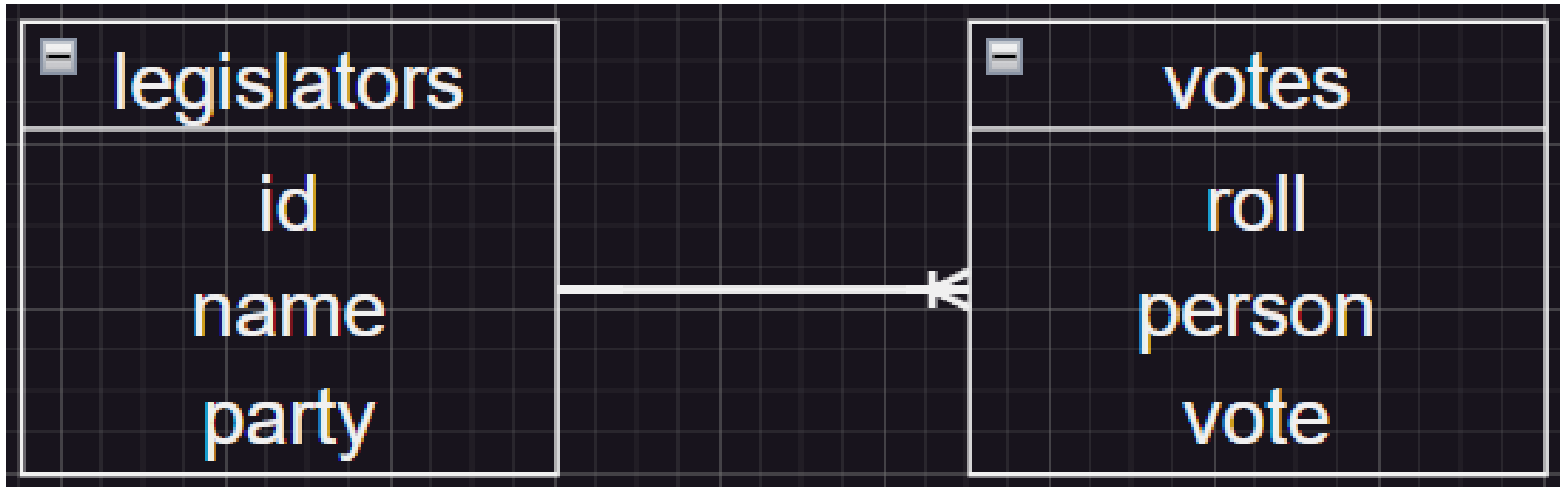
Amendment Agreed to. Simple Majority Required. Source: [senate.gov](#).

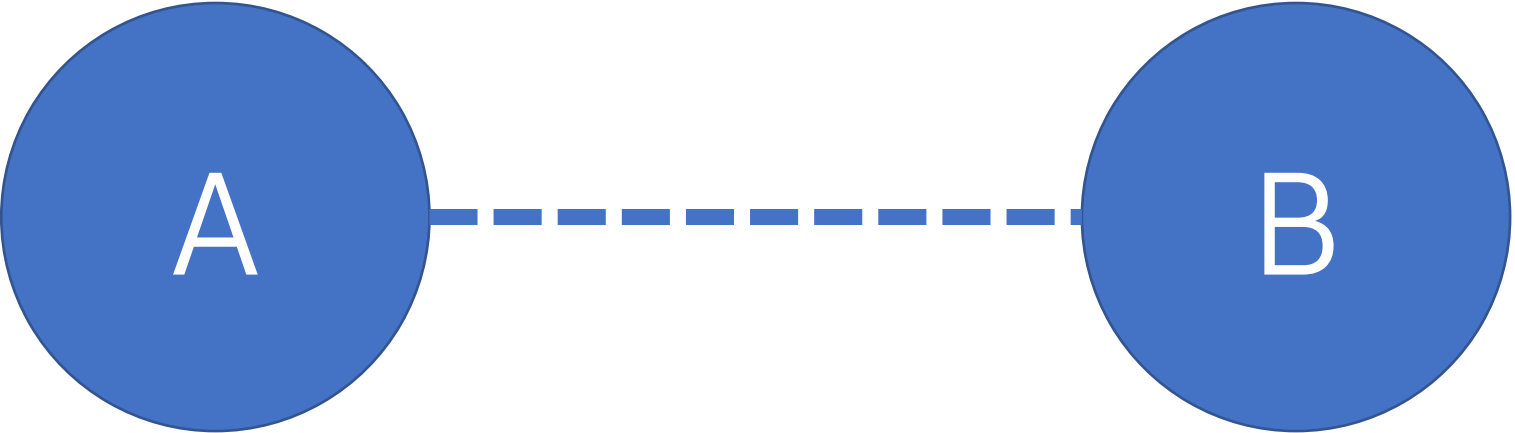
The Yea votes represented 97% of the country's population by apportioning each state's population to its voting senators.

Ideology Vote Chart



Key: Republican - Yea Democrat - Yea
Republican - Nay Democrat - Nay





B

A

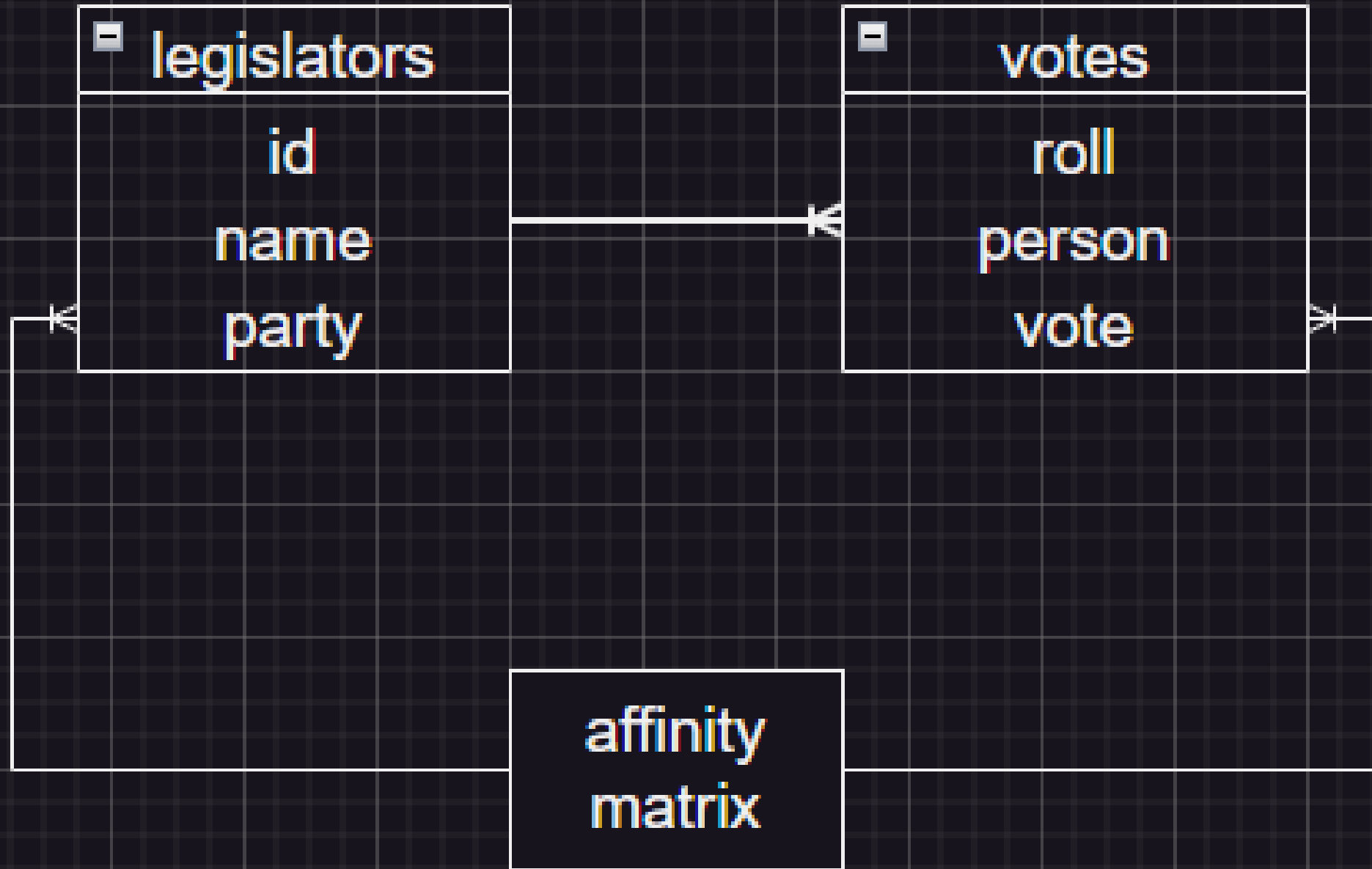
[illegible]

B

[illegible]

$$\frac{f(x) = \sum[A.vote = B.vote]}{n_{rolls}} > 0.65$$

A		B					
			412038	401531	401856	402077	...
		412038	-	(0,1)	(0,2)	(0,3)	(0,n)
		401531	-	-	(1,2)	(1,3)	(1,n)
		401856	-	-	-	(2,3)	(2,n)
		402077	-	-	-	-	(3,n)
		...	-	-	-	-	-



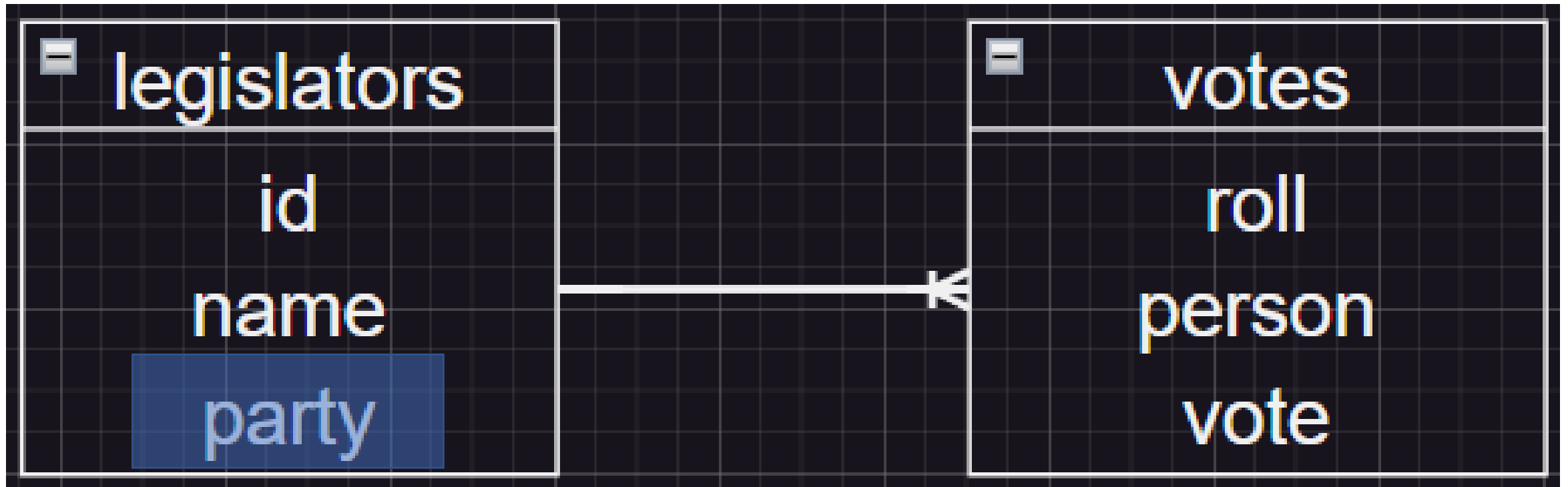
Steps

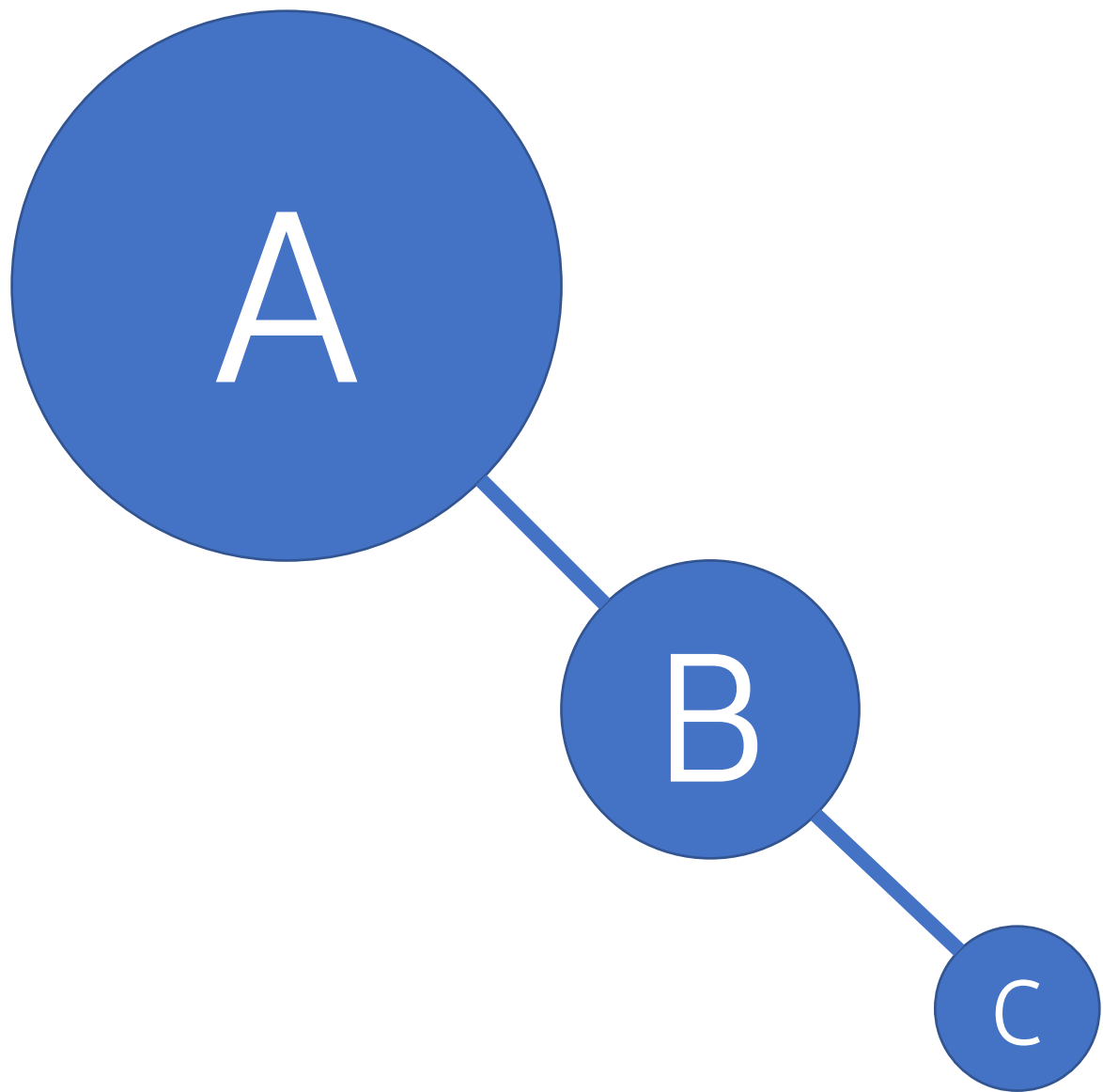
Build a affinity matrix. In codes, we should:

1. Create a symmetric matrix class.
 - $M[i, j] = M[j, i]$
 - What happens inside: Set $M[i, j]$ value only.
2. Define activation function.
3. Build the matrix.

Network diagram elements

- Nodes
 - Colors
 - Sizes
 - Position
 - Name
- Edges
 - Shared colors
 - Shared thicknesses
- Layout





II. Report vs. Dashboard

- Report: Is a static document that stores data as text, table, or visuals.
 - Meaningful for specific time range.
 - Readers cannot interact.
- Dashboard: Visual tools for dynamic data.
 - Real-time data.
 - Interactive.

	Report	Dashboard
State	Static	Dynamic
Data	Past	Present
Features	- Report - Analyze	- Control - Monitor - Dynamic visualization
Strengths	- Reduce noises - Overall analysis - Printing compability	- High visualization - Narrowed analysis
Weaknesses	- Reduced visualization - Ignore current data	- Ignore past data - Synchronization problems - Processing problems - Limited printing